Compact ESP32-S3 Smartwatch PCB with Multi-Sensor Integration

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Project Repository: https://github.com/manishpokhriyal-siliconarch/esp32s3-multisensor-smartwatch

***Abstract*—This paper presents the design and implementation of a compact smartwatch printed circuit board (PCB) based on the ESP32-S3 wireless microcontroller for wearable embedded applications. The proposed hardware platform integrates a MAX30102 photoplethysmography (PPG) sensor for heart-rate monitoring, a BNO086 inertial measurement unit (IMU) for motion sensing, a MEMS microphone for digital audio acquisition, and a capacitive touch display interface for user interaction.**

The smartwatch architecture utilizes I2C, SPI, I2S, USB, Wi-Fi, and Bluetooth communication interfaces for subsystem integration and wireless connectivity. The implemented power-management subsystem incorporates USB Type-C input, MCP73831-based Li-ion battery charging, TPS63001 buck-boost regulation for stable 3.3 V generation, and LD56020 low-dropout regulation for low-noise 1.8 V peripheral supply operation.

The complete schematic and PCB layout were developed using KiCad with emphasis on compact subsystem integration, routing optimization, manufacturability, and wearable mechanical compatibility. The final implementation demonstrates successful realization of a compact two-layer smartwatch PCB integrating sensing, communication, processing, and multi-voltage power-management subsystems within a wearable embedded hardware platform.

***Index Terms*—ESP32-S3, smartwatch PCB, wearable embedded systems, multi-sensor integration, IoT, power management, KiCad, PPG, IMU, MEMS microphone, Bluetooth Low Energy (BLE)**

I. Introduction

Wearable electronics have experienced rapid growth due to increasing demand for compact, low-power, and intelligent embedded systems capable of real-time sensing, wireless communication, and user interaction. Smartwatches have emerged as widely adopted wearable platforms for health monitoring, activity tracking, motion sensing, and portable human-machine interaction. Advancements in embedded processing, wireless communication, and miniaturized sensing technologies have enabled highly integrated wearable systems within constrained physical dimensions.

Despite these advancements, smartwatch hardware design remains challenging due to limitations in PCB area, power consumption, routing complexity, and subsystem integration requirements. Integrating sensing peripherals, communication interfaces, display connectivity, and power-management circuitry within a compact wearable PCB requires careful hardware planning and layout optimization to ensure reliable operation and manufacturability.

This work presents the design and implementation of a compact smartwatch PCB based on the ESP32-S3 wireless microcontroller. The proposed system integrates a MAX30102 photoplethysmography (PPG) sensor for heart-rate monitoring, a BNO086 inertial measurement unit (IMU) for motion sensing, a MEMS microphone for digital audio acquisition, and a capacitive touch display interface for user interaction. Multiple communication protocols including I2C, SPI, I2S, USB, Wi-Fi, and Bluetooth are utilized for subsystem interfacing and wireless connectivity.

The smartwatch architecture was developed using KiCad electronic design automation (EDA) tools with emphasis on compact PCB routing, multi-voltage power distribution, subsystem integration, antenna keep-out considerations, and wearable mechanical compatibility. The implemented power-management subsystem incorporates USB Type-C input, Li-ion battery charging, buck-boost regulation, and low-dropout voltage regulation for stable operation of low-voltage peripherals and embedded processing hardware.

II. System Design

A. Overall System Architecture

The proposed smartwatch platform is centered around the ESP32-S3 wireless microcontroller, which serves as the primary processing, communication, and peripheral-control unit. The implemented architecture integrates sensing, display, audio, wireless communication, and power-management subsystems within a compact two-layer wearable PCB.

The system incorporates a MAX30102 PPG sensor for physiological monitoring, a BNO086 IMU for motion sensing, a MEMS microphone for digital audio acquisition, and a capacitive touch display interface for user interaction. Peripheral interfacing and subsystem communication are achieved using I2C, SPI, I2S, and USB interfaces, while integrated Wi-Fi and Bluetooth connectivity support embedded IoT and wireless wearable applications.

1) Overall Block Diagram: Fig. 1 illustrates the overall system-level architecture of the proposed smartwatch platform including the ESP32-S3 processing subsystem, sensing peripherals, communication interfaces, display subsystem, and power-management circuitry.

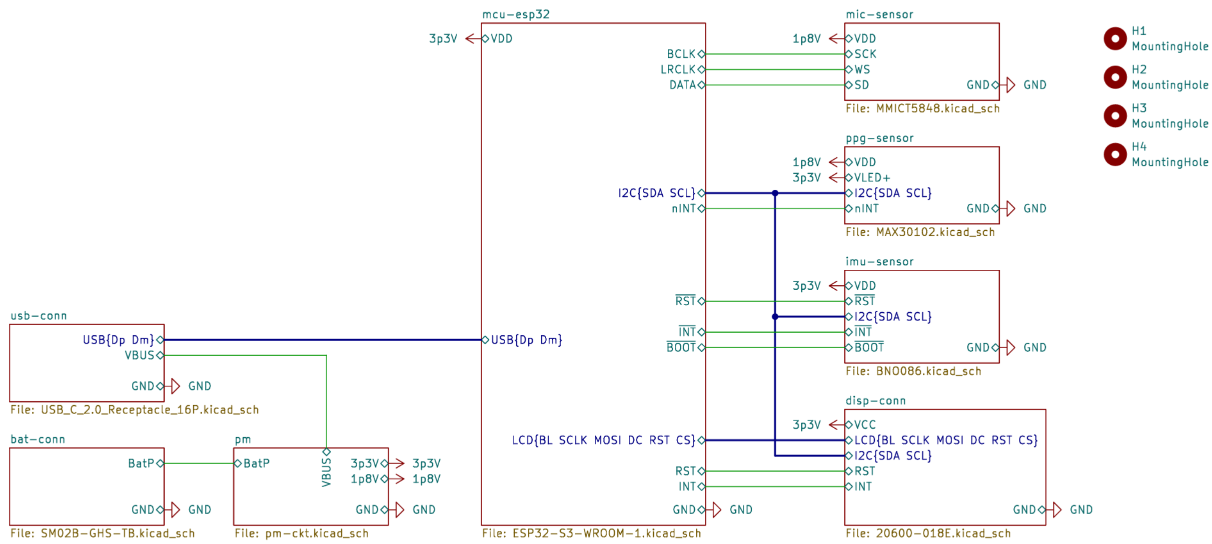

Fig. 1. Top Hierarchical Architectural of the Proposed Smartwatch Platform

B. Hardware Architecture

1) ESP32-S3 Processing Subsystem: The ESP32-S3-WROOM-1-N8R8 module serves as the primary processing and communication unit of the proposed smartwatch platform. The controller was selected due to its compact form factor, integrated Wi-Fi and Bluetooth connectivity, native USB support, onboard PCB antenna, and suitability for low-power wearable embedded applications.

The selected module integrates a dual-core Xtensa LX7 processor operating up to 240 MHz along with 8 MB SPI flash memory and 8 MB PSRAM. The available memory resources support graphical display interfacing, wireless communication stacks, sensor-data buffering, and embedded firmware execution within the smartwatch architecture.

The ESP32-S3 interfaces with sensing peripherals, display circuitry, audio hardware, and power-management subsystems through I2C, SPI, I2S, USB, and UART communication interfaces. Dedicated EN and BOOT switches were additionally incorporated to support manual entry into USB/UART bootloader programming mode during firmware flashing and debugging operations.

The processing subsystem performs sensor acquisition, peripheral communication, embedded processing, wireless data transfer, and display interfacing within the implemented wearable platform.

2) Sensor Integration Subsystem: The sensor integration subsystem incorporates physiological sensing, motion sensing, and digital audio acquisition capabilities required for wearable embedded applications. Multiple sensing peripherals are interfaced with the ESP32-S3 through dedicated communication interfaces to enable reliable sensor-data acquisition, subsystem integration, and real-time wearable operation within the smartwatch architecture.

a) MAX30102 PPG Sensing Subsystem: The MAX30102 photoplethysmography (PPG) sensor was integrated for wearable heart-rate and physiological sensing applications. The device incorporates red and infrared LEDs, integrated photodetectors, analog front-end circuitry, FIFO memory, and an I2C communication interface within a compact low-power wearable-oriented package.

The sensor operates using reflective PPG principles, where variations in reflected optical intensity caused by blood-volume changes are utilized for physiological signal acquisition. The MAX30102 utilizes separate 1.8 V digital/analog and LED supply domains, while communication with the ESP32-S3 is achieved through the I2C interface using the `SDA` and `SCL` signals.

Since the MAX30102 operates at 1.8 V logic levels, a PCA9306 bidirectional level shifter was incorporated between the sensor and ESP32-S3 I2C domains to enable reliable voltage-domain translation while preserving open-drain I2C operation. The sensor additionally provides an active-low interrupt output (`nINT`) for interrupt-driven data acquisition and reduced processor polling overhead.

Dedicated decoupling capacitors and compact routing were incorporated near the sensor supply and communication interfaces to improve power integrity and reduce noise coupling during optical sensing operation.

b) BNO086 IMU Sensing Subsystem: The BNO086 intelligent inertial measurement unit (IMU) was integrated as the primary motion-sensing subsystem of the smartwatch platform. The device incorporates a 3-axis accelerometer, 3-axis gyroscope, and 3-axis magnetometer along with an embedded sensor-fusion engine for real-time motion tracking, orientation estimation, and activity-monitoring applications.

The IMU communicates with the ESP32-S3 through the shared I2C interface using the `SDA` and `SCL` signals. The `PS0/WAKE` and `PS1` configuration pins were connected to ground to enable standard I2C communication mode, while the `CLKSEL0` pin was configured for internal oscillator operation to eliminate the requirement for an external reference clock.

The `INT`, `RST`, and `BOOT` control signals were additionally interfaced with the ESP32-S3 to support interrupt-driven sensing, hardware reset, and firmware-management operations. The interrupt-driven acquisition mechanism reduces continuous processor polling and improves overall wearable-system power efficiency.

The BNO086 operates from the shared 3.3 V supply rail, while local decoupling capacitors were incorporated near the supply pins to improve power integrity and reduce noise during motion-sensing operation.

c) MEMS Microphone Subsystem: The MMICT5848 digital MEMS microphone was integrated as the audio-acquisition subsystem of the smartwatch platform. The device was selected due to its compact package size, low-power operation, integrated digital audio processing, and direct I2S interface support suitable for wearable embedded applications.

The microphone incorporates a MEMS sensing element, analog front-end, ADC, digital filtering circuitry, and an I2S-compatible digital output interface within a compact surface-mount package. Communication with the ESP32-S3 is achieved through the I2S interface using the `SCK`, `WS`, and `SD` signals, where the ESP32-S3 operates as the I2S master and provides clock and frame-synchronization signals.

The microphone operates from the dedicated 1.8 V supply rail with local decoupling capacitance incorporated near the supply pin to improve power integrity during audio acquisition. The `LR` pin was connected to ground to configure left-channel I2S operation, while the implemented smartwatch architecture primarily utilizes the microphone in low-power 16-bit audio mode for energy-efficient wearable sensing applications.

A 100 kΩ pull-down resistor was additionally connected to the `SD` output line to prevent floating of the shared I2S data bus when the microphone output enters a high-impedance state during inactive transmission intervals.

The `THSEL` and `WAKE` pins associated with Acoustic Activity Detection (AAD) functionality were intentionally left unconnected since voice-triggered wake functionality is not utilized in the current smartwatch implementation.

3) Power Management Subsystem: The smartwatch platform utilizes a rechargeable Li-ion battery power architecture designed for stable multi-voltage wearable operation. The implemented subsystem incorporates USB Type-C power input, MCP73831-based Li-ion battery charging, TPS63001 buck-boost regulation for stable 3.3 V generation, and LD56020 low-dropout regulation for low-noise 1.8 V peripheral supply operation.

The multi-stage power architecture supports embedded processing, wireless communication, sensing peripherals, and display circuitry while maintaining compact subsystem integration within the constrained wearable PCB layout.

a) USB Type-C Power and Programming Interface: The smartwatch platform incorporates a USB Type-C interface for external power input, battery charging, firmware flashing, and debugging operations. The connector operates in USB 2.0

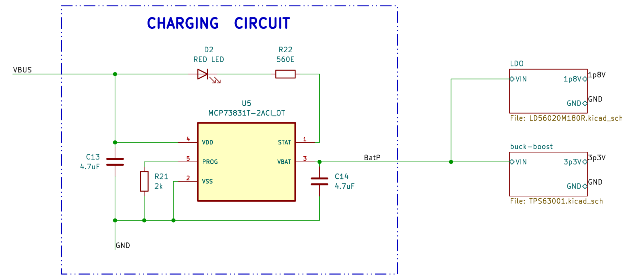

Fig. 2. Power Management Architecture of the Smartwatch

mode, where the `D+` and `D−` differential lines are connected directly to the native USB interface of the ESP32-S3 for programming and serial communication.

The `VBUS` pin provides the primary 5 V input supply for the power-management subsystem and battery-charging circuitry. The `CC1` and `CC2` configuration-channel pins are connected to ground through 5.1 kΩ pull-down resistors to configure the smartwatch as a USB sink device and support reversible cable-orientation detection.

The `SBU1` and `SBU2` sideband pins remain unused since alternate-mode functionality is not required in the implemented smartwatch architecture.

b) Li-ion Battery Charging Subsystem: The battery-charging subsystem is implemented using the `MCP73831-2ACI/OT` linear charge-management controller for single-cell Li-ion battery charging. The device supports constant-current/constant-voltage (CC/CV) charging operation from the USB Type-C input supply.

The USB `VBUS` line is connected to the `VDD` input of the MCP73831, while 4.7 μF capacitors were incorporated at the `VDD` and `VBAT` pins to improve input and output stability. The `VBAT` pin is connected directly to the positive battery terminal.

The charging current is configured using a resistor connected between the `PROG` and `VSS` pins, where a 2 kΩ resistor sets the charging current close to 500 mA. The `STAT` output drives a red LED through a current-limiting resistor to provide visual charging-status indication during active charging operation.

The MCP73831 additionally provides integrated thermal regulation, automatic recharge, undervoltage lockout, and reverse-discharge protection for safe rechargeable battery operation within the smartwatch platform.

c) 3.3 V Buck-Boost Power Regulation Subsystem: The primary regulated 3.3 V supply rail of the smartwatch platform is generated using the `TPS63001` high-efficiency buck-boost DC-DC converter. The regulator was selected due to its compact implementation, low quiescent current, and ability to maintain stable output regulation under varying Li-ion battery voltages.

The TPS63001 supports automatic transition between buck and boost operating modes, enabling stable 3.3 V generation during battery charge and discharge conditions. In the implemented design, the battery supply is connected to the `VIN` and `VINA` inputs, while input and output filtering are achieved using ceramic capacitors to improve transient response and reduce supply ripple.

A 2.2 μH inductor connected between the `L1` and `L2` switching nodes supports synchronous buck-boost conversion. Since the fixed 3.3 V regulator variant is utilized, the `FB` pin is connected directly to the regulated output node.

The `EN` pin is configured for continuous regulator operation during normal system activity, while separate `GND` and `PGND` connections reduce switching-noise coupling and improve regulator stability. The TPS63001 additionally incorporates synchronous rectification, soft-start control, thermal protection, undervoltage lockout, and automatic power-save functionality for efficient low-power wearable operation.

d) 1.8 V LDO Regulation Subsystem: A dedicated 1.8 V supply rail is implemented using the fixed-output `LD56020M180R` low-dropout (LDO) voltage regulator. The regulator provides a stable low-noise supply for low-voltage peripherals including the MAX30102 sensing subsystem and MEMS microphone circuitry. The LD56020 was selected due to its low dropout voltage, low quiescent current, compact footprint, and suitability for sensitive wearable embedded applications.

The regulator input is bypassed using a 1 μF capacitor connected near the `VIN` pin, while an additional 1 μF capacitor is connected at the `VOUT` pin to improve output stability and transient response. Since the implemented device is a fixed-output variant, no external feedback network is required.

The `EN` pin is tied high for continuous operation during normal system activity, while the regulator additionally incorporates thermal shutdown, short-circuit protection, and electronic shutdown functionality for reliable low-power operation within the smartwatch platform.

e) 1.83-inch Touch LCD Display Module: A 1.83-inch IPS touch LCD module was selected as the primary display subsystem of the smartwatch platform due to its compact smartwatch-compatible form factor, integrated capacitive touch functionality, and compatibility with ESP32-based embedded systems.

Fig. 3. Rear-side view of the 1.83-inch touch LCD module showing the selected 18P FPC connector used for compact and reliable interfacing.

The display provides a resolution of 240 × 284 pixels with 262K color support using an IPS panel architecture for improved viewing angles and display clarity. The module dimensions of approximately 35 mm × 42 mm directly influenced the PCB outline and mechanical layout of the smartwatch platform.

The module integrates an ST7789P display-driver IC utilizing SPI communication for graphical display interfacing and a CST816D capacitive touch controller utilizing the I2C interface for touch-event acquisition. Display and touch-control signals are interfaced directly with the ESP32-S3 processing subsystem for display rendering, touch detection, reset control, and backlight operation.

The display module interfaces with the smartwatch PCB through a rear-mounted FPC connector, while the PCB geometry and mounting-hole arrangement were designed to align with the selected display-module dimensions for direct wearable mechanical integration.

C. Component Selection

TABLE I
MAJOR HARDWARE COMPONENTS USED IN THE SMARTWATCH

Component	Manufacturer	MPN
Wireless MCU	Espressif Systems	ESP32-S3-WROOM-1-N8R8
PPG Sensor	Analog Devices	MAX30102EFD+T
IMU Sensor	CEVA Technologies	BNO086
MEMS Mic	TDK InvenSense	MMICT5848-00-012
Battery Charger IC	Microchip Technology	MCP73831T-2ACI/OT
Buck-Boost Converter	Texas Instruments	TPS63001DRCR
1.8 V LDO	STMicroelectronics	LD56020M180R
I2C Level Shifter	NXP Semiconductors	PCA9306DP1,125
USB Type-C Connector	GCT	USB4105-GF-A
Display Connector	I-PEX	20600-018E-01
Battery Connector	JST	SM02B-GHS-TB
1.83-inch LCD Module	Waveshare	1.83inch-Touch-LCD-Module

D. Communication Interfaces

Multiple communication protocols are implemented within the smartwatch architecture for peripheral interfacing, subsystem communication, and wireless connectivity.

The I2C interface is utilized for communication between the ESP32-S3 and multiple peripherals including the MAX30102 PPG sensor, BNO086 IMU sensor, and capacitive touch controller. The shared I2C architecture minimizes GPIO utilization and routing complexity within the compact wearable PCB layout.

Digital audio acquisition is implemented using the I2S interface between the ESP32-S3 and the MEMS microphone subsystem through dedicated clock, word-select, and serial-data signals. The display subsystem utilizes an SPI interface for high-speed graphical data transfer between the ESP32-S3 and the display-driver circuitry.

The ESP32-S3 additionally provides integrated Wi-Fi and Bluetooth connectivity for wireless communication, embedded IoT interfacing, and smartwatch-to-smartphone connectivity applications.

E. PCB Layout Design

The PCB layout was developed using KiCad as a compact two-layer wearable embedded hardware platform integrating sensing, processing, communication, display, audio, and power-management subsystems. The layout design emphasized compact component placement, efficient routing, stable power distribution, manufacturability, and wearable mechanical compatibility within constrained smartwatch dimensions. Standard 0402 passive components were utilized for compact PCB integration, while 0603 LEDs improved visual indication and assembly reliability. A 1008 (2520 Metric) SMD inductor was additionally selected for the TPS63001 power stage.

1) Component Placement Strategy: Component placement was optimized to minimize routing complexity, reduce interconnect lengths, and improve subsystem organization within the constrained smartwatch PCB area.

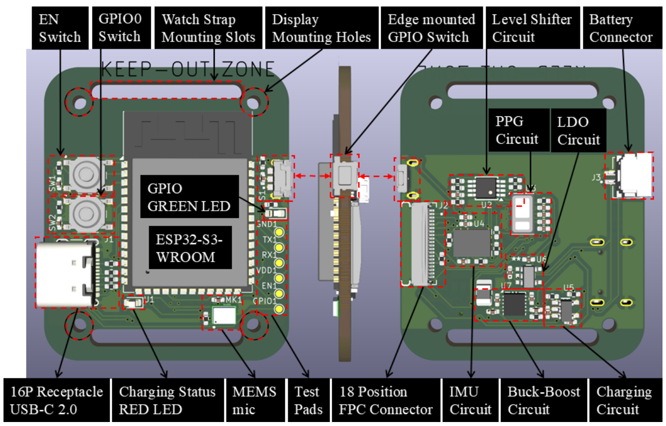

Fig. 4. Annotated component placement layout of the smartwatch PCB.

The smartwatch PCB was implemented as a compact two-layer wearable platform with overall dimensions of approximately 35 mm × 42 mm based on commercially available 1.83-inch smartwatch display modules. The PCB geometry and mounting arrangement were designed to align with the selected display-module dimensions for simplified wearable mechanical integration.

The front-side PCB layout primarily integrates the ESP32-S3 processing subsystem, USB Type-C interface, MEMS microphone, user-control switches, side-mounted smartwatch control button, programming interface, and debugging test pads. The rear-side layout incorporates the MAX30102 PPG sensor, BNO086 IMU, display FPC connector, charging circuitry, buck-boost converter, LDO regulator, level-shifter circuitry, and battery connector.

The ESP32-S3 module was positioned beneath a dedicated antenna keep-out region to preserve wireless communication performance, while sensing peripherals were placed close to their associated interfaces to reduce routing congestion and improve signal integrity. The rear-mounted FPC connector enables compact integration of the display module above the PCB structure.

The MAX30102 PPG subsystem was positioned near the center of the rear PCB surface to ensure direct skin alignment during wearable operation and improve physiological signal acquisition reliability. The battery connector was placed along the rear PCB region to support external Li-Po battery attachment while maintaining clearance around the optical sensing area.

Power-management circuitry was grouped together to establish a compact power-delivery path between the USB input, charging subsystem, battery interface, buck-boost regulator, and LDO stage. Dedicated test pads were additionally incorporated near critical communication, power, and programming interfaces to support debugging and hardware verification.

Watch-strap mounting slots of approximately 22 mm × 2.5 mm were integrated into the PCB outline to support standard smartwatch strap attachment and wearable mechanical assembly.

2) Routing and Ground Plane Design: PCB routing was performed after component-placement optimization to establish efficient interconnection between the processing, sensing, communication, display, and power-management subsystems. Due to the constrained wearable form factor, the smartwatch was implemented as a compact two-layer PCB to balance routing complexity, manufacturability, and subsystem integration.

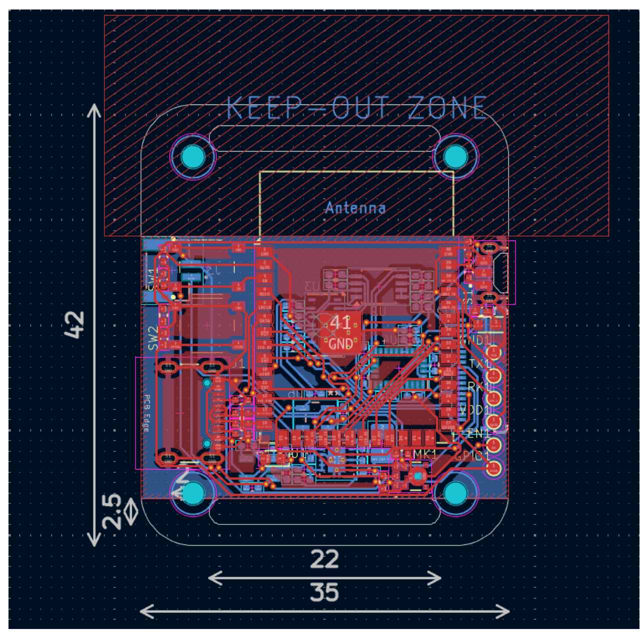

Fig. 5. PCB Routing and Ground Plane Layout

Routing was organized to minimize overall trace length, reduce routing overlap, and maintain compact signal distribution across the smartwatch architecture. Sensitive interfaces including I2C, I2S, SPI, and USB were routed using short interconnect paths to improve communication reliability and reduce interference coupling between adjacent subsystems.

Both PCB layers were utilized collaboratively for signal routing and subsystem interconnection while satisfying area-constrained wearable layout requirements. Continuous ground-

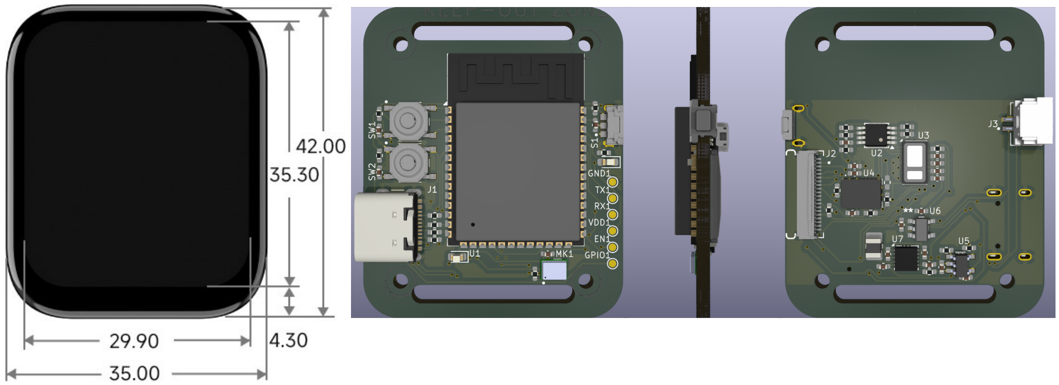

Fig. 6. Final smartwatch PCB implementation showing dimensional comparison with the selected 1.83-inch smartwatch display module, front-side PCB layout, edge-view visualization, and rear-side PCB layout. The PCB outline and display mounting-hole arrangement were designed to align with the selected display module, enabling direct mounting of the display above the front-side PCB structure.

plane regions and copper pours were additionally incorporated to improve return-current continuity, subsystem stability, and electromagnetic interference suppression. Local decoupling capacitors were placed near integrated circuits and power-input pins to improve supply stability and transient response.

3) Design Rule Verification (DRC/ERC): Electrical Rule Check (ERC) and Design Rule Check (DRC) verification were performed using the verification tools available in KiCad 9.0.7 to validate schematic connectivity, routing integrity, PCB clearances, footprint association, and manufacturability compliance.

The ERC process verified subsystem connectivity, power-net association, and schematic-level interconnection integrity across the implemented hardware architecture. Similarly, DRC verification was performed after PCB routing completion to validate trace clearances, copper spacing, via placement, and layout-rule compliance for reliable PCB fabrication.

The implemented smartwatch PCB satisfied all major ERC and DRC verification requirements without critical violations affecting subsystem functionality or manufacturability. The only reported condition corresponds to mechanical mounting holes positioned within the ESP32-S3 antenna keep-out region. Since these structures are non-conductive mechanical features, the condition does not introduce practical interference to wireless operation within the implemented smartwatch platform.

III. Results

The proposed smartwatch platform was successfully implemented as a compact two-layer wearable PCB integrating sensing, processing, communication, display interfacing, and multi-voltage power-management subsystems within a constrained smartwatch form factor.

Successful hardware integration was achieved for the ESP32-S3 processing subsystem, MAX30102 PPG sensor, BNO086 IMU sensor, MEMS microphone subsystem, display interface circuitry, and rechargeable power-management architecture. The implemented design additionally integrates I2C, I2S, SPI, USB, Wi-Fi, and Bluetooth communication interfaces required for smartwatch operation.

The smartwatch PCB was implemented with overall dimensions of approximately 35 mm $\times$ 42 mm and incorporates display mounting holes, watch-strap mounting slots, side-mounted operational control switch, and rear-side sensing integration suitable for wearable applications. The MAX30102 PPG sensing subsystem was positioned near the center of the rear PCB surface to improve skin alignment during physiological signal acquisition.

Stable regulated 3.3 V and 1.8 V supply rails were achieved using the TPS63001 buck-boost converter and LD56020 LDO regulator respectively. USB Type-C connectivity and native USB interfacing with the ESP32-S3 were additionally integrated for external power input, firmware flashing, debugging, and USB device detection.

Mechanical feasibility of the smartwatch platform was validated through compact PCB geometry, display-aligned board dimensions, rear-mounted display interfacing, and organized battery-placement integration compatible with smartwatch assembly requirements.

ERC and DRC verification were successfully completed using KiCad 9.0.7 verification tools. The implemented schematic and PCB layout satisfied major connectivity, clearance, routing, and manufacturability requirements without critical violations affecting subsystem functionality or PCB fabrication feasibility.

The final implementation demonstrates successful realization of a compact wearable smartwatch PCB integrating sensing, communication, processing, and power-management subsystems within a manufacturable embedded hardware platform.

IV. Conclusion

This work presented the design and implementation of a compact smartwatch PCB integrating sensing, processing, communication, display interfacing, and rechargeable power-management subsystems within a wearable form factor. The proposed hardware platform was successfully developed using KiCad 9.0.7 with emphasis on compact subsystem integration, reliable peripheral interfacing, manufacturable PCB layout design, and stable multi-voltage operation.

Successful integration was achieved for the ESP32-S3 wireless processing subsystem, MAX30102 PPG sensor, BNO086 IMU sensor, MEMS microphone subsystem, display interface circuitry, USB Type-C connectivity, and Li-ion battery-management architecture. The implemented smartwatch PCB additionally incorporates watch-strap mounting slots, display mounting support, debugging interfaces, and a side-mounted operational control button similar to conventional smartwatch platforms for user interaction and device control.

Stable regulated 3.3 V and 1.8 V supply rails were achieved using the TPS63001 buck-boost converter and LD56020 LDO regulator respectively. ERC and DRC verification further confirmed schematic connectivity, routing integrity, and manufacturability compliance for the implemented hardware architecture.

The final smartwatch PCB demonstrates successful realization of a compact two-layer wearable embedded platform suitable for physiological sensing, motion tracking, digital audio acquisition, wireless communication, and embedded IoT applications. Future work may include firmware optimization, enclosure development, battery-life improvement, and BLE-only low-power operation for smartwatch-to-smartphone wireless connectivity and sensor-data synchronization.

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